Week 1 web page: delivery guide

Instructor notes for Week1_IntroAndAlgorithms.html

General

- Page: https://kevintshoemaker.github.io/NRES-746/Week1_IntroAndAlgorithms.html. Open it before class; the three widgets need no internet once loaded.
- For live demos, run the matching section of `CODE_Week1_IntroAndAlgorithms.R` (same seeds, same numbers). The 100,000-sample salmon loop takes under a second.
- Zoom the browser to 125-150% for the projector.
- Questions on the page have no answers. Answers are below; *Land:* is where discussion should end up.
- Readings for Canvas: Bolker ch. 1 (printed pp. 1-35; 1.3.2 on p. 9 is the analytical vs. computational distinction) and appendix A (A.1-A.4, pp. 480-482). The schedule also lists ch. 2-3.
- WODs: *Math & Stats Self-Diagnostic* (Monday) and *Pseudocoding a Permutation Test* (Wednesday), keys in `wod/answer_keys/`. Both are worked on the page (`#wod-diagnostic`, `#wod-permutation`).
- Lab 1 Act answers stay off the page. The lab section avoids the linearization of ae^{-bx} (Act 2, 1c), the Ricker derivative and critical point (Act 2, Part 2), the 0-to-2 ramp (Act 2, Part 3), and SSR (Act 3). Its examples use different functions and numbers: a power law, x^2e^{-x}, a 0-to-4 ramp over 5 days.

As taught (Aug. 24-26, 2026)

- The page was built Nov. 1, after the week, from the legacy `INTRO` and `LECTURE1` pages, `NOTES_Week1_Lecture2_AlgorithmsInStatistics`, `NOTES_Lab1_MathRefresher` (both now in `lecture_notes/archive/`), the two WODs, Lab 1, and Bolker ch. 1 and appendix A.
- Monday was the syllabus and the diagnostic WOD only, no content.
- Wednesday: the workflow big picture was planned but not covered live (it opened Week 2 instead). The page keeps a short version in the first H1 as a course map. The bootstrap was a cut-first bonus; no record of whether it was reached.
- New on the page: the three widgets, the power-law example, the course-logistics section, the discussion prompts, and the packaged `permutation_test()` (the legacy version read global variables instead of its own arguments).
- `LECTURE1` and `INTRO` are archived now that this page covers them.

Part 1: What this course is about (Monday, ~15-20 min after the syllabus)

Big picture (~5 min)

- Walk the workflow left to right: problem first, then theories, then models (function + noise), fit, compare, revise. Point out where the semester's topics sit (functions, probability, simulation = MODELS; likelihood, Bayes, MCMC = FIT; AIC, CV = COMPARE).
- "Playing" with data, models, and probability: simulate from your own model and see if you can recover the parameters.

How the course runs (~3 min)

- WOD routine, Acts (alone, then group; wrong first attempts are expected), final project pitches, AI policy (closed in class unless a handout says otherwise), "stop me if it doesn't make sense."

WOD (~10 min, including the neighbor comparison)

- Answers: $x = 5$; $r = \ln(5.5)/8 = 0.213$; $f'(x) = 5(4x^2 - 3x)^4(8x - 3)$, $f'(1) = 25$; $R(t) = 2t^2$, total 50; $P(A \cup B) = 0.7$; two children 1/3; CI (b); p-value (b); `site_data[, 2]`; `for (i in seq_along(values))`; braces around the function body.
- Collect the sheets and look at the self-ratings before Lab 1. Low calculus ratings mean more time on the lab's derivative section.

Discussion (~3-5 min)

- Q1 (CI of 0.42 to 0.61, "95% chance the truth is in there") *Land:* strictly, the 95% belongs to the procedure: in repeated samples, 95% of such intervals cover the true value; this one either does or doesn't. A Bayesian credible interval does carry that reading (given a prior), which comes later in the semester. For most management decisions the distinction doesn't change the action; it matters when someone treats the interval's ends as hard limits.
- Q2 (your own problem and two theories) *Land:* open. Push for a problem stated as a decision or need, and for theories that predict different data; if they predict the same data, no study separates them.

Part 2: Math tools for Lab 1 (lab opening, ~10-15 min)

- A refresher, not a first lesson. Do logs and the product + chain rules; skim integration if time is short (Act 2 Part 3 gives a worked constant-rate example on the handout).
- **Board:** exponent rules, then logs as "what power?"; the growing square, then the rectangle.
- Q (power law: rise per tenfold mass; slope of 1) *Land:* 0.75 per decade (0.477, 1.227, 1.977, 2.727). A slope of 1 means rate is proportional to mass, so the rate per gram is

constant; a slope of 0.75 means larger animals use less energy per gram.

- **Widget (growing rectangle):** square mode, $x = 3$. At $dx = 1.5$: exact change 11.25, strips 9, corner 2.25 (20%). At $dx = 0.1$: exact 0.61, strips 0.60, corner 0.01 (1.6%). Ask for a prediction before sliding. Then rectangle mode: shrink h and watch the corner share go to 0.
- Q (corner's share; why drop it) *Land:* the share is $dx/(2x + dx)$, which goes to 0. Divided by dx, the corner contributes dx, which vanishes in the limit, so it doesn't affect the derivative. The exact new area still includes it.
- x^2e^{-x}: $f' = x(2 - x)e^{-x}$, critical points 0 and 2. This is the closest parallel to point stuck groups toward in Act 2 Part 2, without giving the Ricker answer.
- **Widget (accumulation):** start with the constant rate (any rule is exact, area 10). Ramp, left edge, 5 rectangles: 8; right edge: 12; midpoint: 10. Rise-and-fall, left edge, 5 rectangles: 6.38 vs. 7.13 exact; midpoint 7.24. Slide to 100 rectangles.
- Q (left/right/midpoint on the ramp) *Land:* the ramp increases, so the left edge takes the lowest height in each slice (too small) and the right edge the highest (too big). For a straight line, the midpoint's overshoot on one half of each slice exactly cancels the undershoot on the other; a curved function doesn't cancel.
- Circulating: Act 1 and Act 3 need none of this. Act 2 Part 1 needs log rules; Part 2 product + chain; Part 3 "area = total."

Part 3: Algorithms for inference (Wednesday, ~32 min)

Big picture (~3 min)

- COMPARE box, simplest version: data vs. a null model. Analytical (formula, exact, needs assumptions) vs. computational (a program, approximate, needs only the story). Bolker 1.3.2.

WOD (~6 min)

- Statistic: difference in means, $174.8 - 178.2 = -3.4$ g. Two-tailed (no direction predicted). Pseudocode: shuffle labels, recompute, store, compare. Have a couple of students read their steps aloud before showing the page's list; the R version waits for section 4.

Two routes to a z-test (~10 min)

- $\bar{x} = 3.476$, $SE = 0.285$, $z = -3.60$, $p = 0.00016$.
- Simulation: 21 of 100,000 means at or below 3.476, $p \approx 0.00021$. SD of simulated means 0.286 vs. $SE = 0.285$: the loop finds the standard error without the formula.
- **Board pseudocode:** simulate a sample under the null, compute its mean, store; repeat; p = fraction at least as extreme.
- One-tailed here vs. two-tailed for the lizards; worth a check-in.

Permutation test (~10 min)

- $p \approx 0.37$ (seed 746); t-test 0.36. Not significant, a useful contrast with the salmon.
- Put the page's loop next to the WOD's fill-in skeleton.
- **Widget (resampling):** lizards first, 10,000 replicates (about 0.37, the shaded tails on both sides). Then salmon: add 1,000 at a time; most runs show 0 extreme means. Ask what p to report before answering.
- Q (salmon runs of 1,000 with no extreme means) *Land:* report $p < 0.001$ (at most 1 in 1,000), not $p = 0$. With true $p = 0.00016$, 1,000 replicates expect 0.16 hits, so about 85% of runs show none. Seeing about 5 takes roughly 30,000 replicates; the page's 100,000 gave 21.
- `permutation_test()`: packaging the loop. Skip if behind.

Discussion (~3-5 min)

- Q1 (formula 0.00016 vs. simulation 0.00021) *Land:* the simulation's own standard error is about $\sqrt{p(1-p)/N} = 0.00004$, so they agree within noise. Trust the formula when its assumptions hold (here they are given); trust the simulation when the formula's assumptions fail or no formula exists, and add replicates until the noise is small relative to the decision.
- Q2 (when permutation and t-tests disagree) *Land:* small samples with strong skew or an outlier (the t-test leans on normality of the mean), or very unequal spreads. The permutation test assumes only that labels are exchangeable under the null, so it holds up there.
- Q3 (own research) *Land:* open. The random step is whatever the null says is arbitrary: labels, locations, timing, or draws from a simulated process.

Bonus (cut first): bootstrap

- R^2: girth 0.94, height 0.36. 95% bootstrap intervals: girth 0.90 to 0.96, height 0.14 to 0.59. Third flavor of the same loop (resample rows with replacement). Lab 2 Act 3 uses a bootstrap, so don't work its example here.